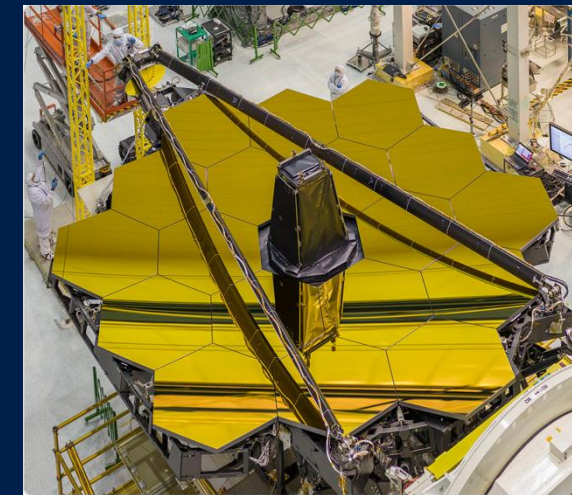


# Exploring ML Models for Detection of Atmospheric Composition in Exoplanets

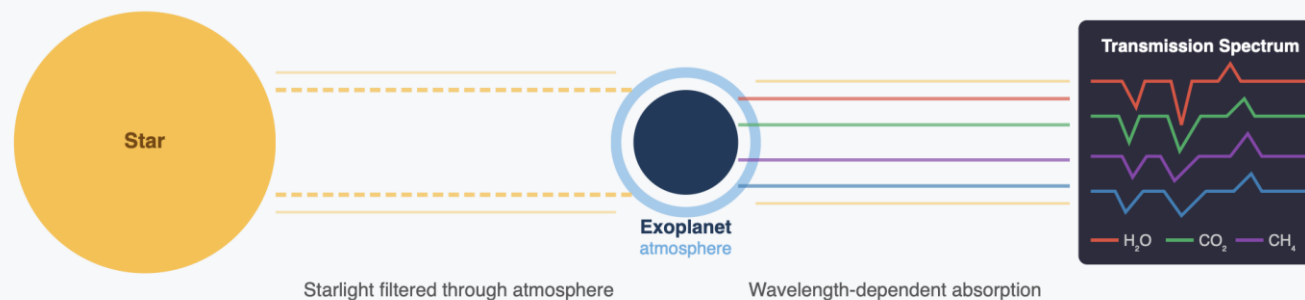
*Machine learning–based atmospheric retrieval from transmission spectra*

Dataset: INARA (NASA FDL / Zorzan et al. 2025)



## Team Members

Shantanu Wankhare wankhare.s@northeastern.edu  
Bhalchandra Shinde shinde.b@northeastern.edu  
Asad Mulani mulani.a@northeastern.edu

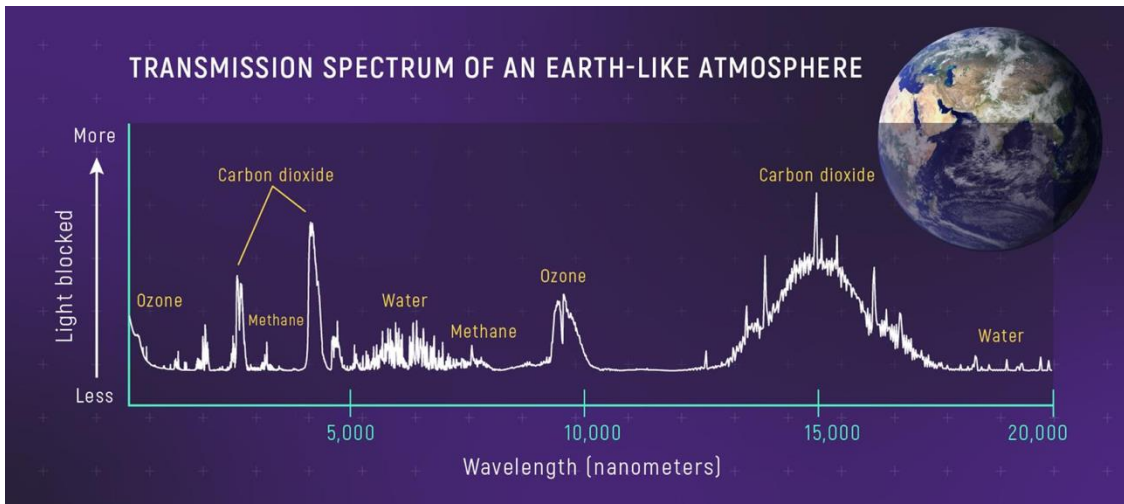


# Motivation — Why This Matters

The intersection of scientific urgency and methodological opportunity

## The Science

- ▶ Rocky exoplanet atmospheres reveal if a world could be habitable
- ▶ Co-detection of  $O_2 + CH_4$  is a strong biosignature — signals biological activity
- ▶ JWST + planned Habitable Worlds Observatory (HWO) produce spectra at unprecedented scale
- ▶ Population-level surveys require analysing thousands of planets simultaneously



## Traditional Bayesian Retrieval

- ✗ Nested sampling / MCMC — rigorous but slow
- ✗ Hours to days per planet
- ✗ Does not scale to thousands of targets

## ML-Based Retrieval

- ✓ Milliseconds per planet once trained
- ✓ Enables population-level atmospheric studies
- ✓ Active research direction (JWST era, 2022–present)
- ✓ Established benchmarks exist (Márquez-Neila 2018; Vasist 2023; Gebhard 2024)

# Scientific Context — JWST & Transit Spectroscopy

From telescope observations to ML-powered atmospheric retrieval

## Transit Spectroscopy

- ▶ Planet passes in front of its host star (transit event)
- ▶ Starlight filters through the planet's atmosphere
- ▶ Each molecule absorbs specific wavelengths → leaves a spectral fingerprint
- ▶ Transmission spectrum encodes atmospheric chemical composition
- ▶ Benchmark: WASP-39 b (JWST, Nature 614:649, 2023) — confirmed CO<sub>2</sub>, SO<sub>2</sub>, H<sub>2</sub>O, CO in a single observation

*Star • → [Planet transit] → Spectrograph → Absorption dips per molecule*

## Why JWST Is a Game-Changer

- ▶ NIRSpec (0.6–5.3 μm) + MIRI (5–28 μm) — complete spectral coverage
- ▶ Covers nearly every key biosignature: O<sub>2</sub>, O<sub>3</sub>, CH<sub>4</sub>, H<sub>2</sub>O, CO<sub>2</sub>, N<sub>2</sub>O, SO<sub>2</sub>
- ▶ Launched 25 Dec 2021 — science operations began 2022
- ▶ Will survey thousands of exoplanet atmospheres over its lifetime
- ▶ Population-level analysis demands millisecond-per-planet ML retrieval

## INARA → JWST Pipeline

- ▶ INARA: first large-scale ML benchmark for atmospheric retrieval (Zorzan et al. 2025)
- ▶ Synthetic CLIMA profiles (12 × 101) → same 12 molecules JWST observes
- ▶ 3.1 million simulations covering F, G, K, M stellar types
- ▶ Future: apply trained model directly to real JWST transmission spectra
- ▶ PSG secondary dataset (3 ch × 4,378 pts, 0.2–2.0 μm) ready for HWO spectra

# Problem Statement

Multi-output regression over molecular  $\log_{10}$  abundances

Given a transmission spectrum of an exoplanet's atmosphere,  
**predict the  $\log_{10}$  surface volume mixing ratios of 12 molecular species.**

## INPUT

### Transmission Spectrum

12 CLIMA channels  
101 altitude levels  
Z-score normalised  
(INARA ATMOS format)



## ML MODEL

### Random Forest

*(baseline,  $\leq 10k$  samples)*

### 1D CNN

*(deep model, full dataset)*



## OUTPUT

### 12 Molecular Abundances

$\text{H}_2\text{O} \cdot \text{CO}_2 \cdot \text{O}_2 \cdot \text{O}_3$   
 $\text{CH}_4 \cdot \text{N}_2 \cdot \text{N}_2\text{O} \cdot \text{CO}$   
 $\text{H}_2 \cdot \text{H}_2\text{S} \cdot \text{SO}_2 \cdot \text{NH}_3$   
( $\log_{10}$  VMR at surface)

*Why  $\log_{10}$  space? Mixing ratios span  $\sim 12$  orders of magnitude ( $\sim 0.1$  for  $\text{N}_2$  to  $10^{-40}$  for trace species).  $\log_{10}$  makes the prediction scale uniform — an error of 1.0 means 'off by one order of magnitude'.*

# Dataset — INARA (NASA FDL / Zorzan et al. 2025)

3.1 million synthetic rocky planet spectra from NASA Planetary Spectrum Generator

## Dataset Facts

|                   |                                                  |
|-------------------|--------------------------------------------------|
| Full dataset size | 3,112,620 synthetic CLIMA profiles               |
| This project uses | 25,000 samples (local dev run; full 3.1M on HPC) |
| Profile type      | INARA ATMOS — CLIMA atmospheric profiles         |
| Altitude levels   | 101 per profile                                  |
| Input channels    | 12 CLIMA variables (T, P, 10 molecules)          |
| Stellar types     | F, G, K, M — 4 types                             |
| Format            | CSV per planet → .npy arrays after processing    |

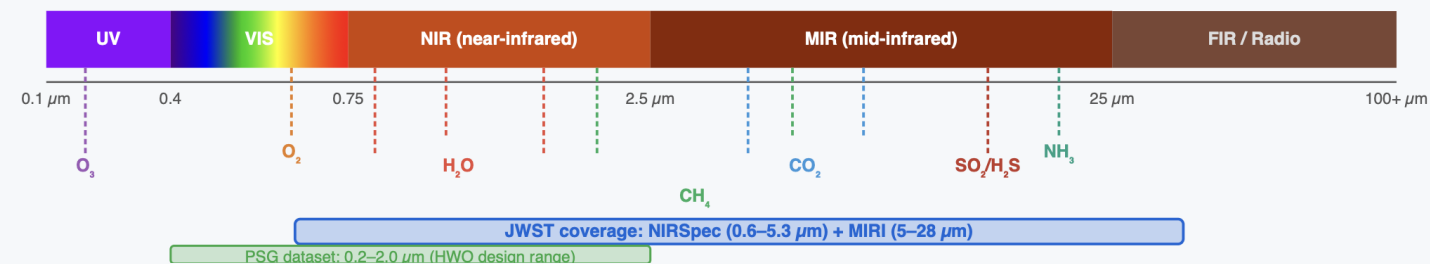
## CLIMA Input Channels (12)

|        |                                                                                                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T / P  | Temperature & Pressure profiles — primary physical state of the atmosphere                                                                                                                         |
| 10 mol | H <sub>2</sub> O, CO <sub>2</sub> , O <sub>2</sub> , O <sub>3</sub> , CH <sub>4</sub> , N <sub>2</sub> , N <sub>2</sub> O, CO, H <sub>2</sub> , H <sub>2</sub> S — altitude-resolved mixing ratios |
| Nor m. | All 12 channels Z-score normalised per channel (fit on train only)                                                                                                                                 |

## 12 Target Molecules

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| Habitability:  | H <sub>2</sub> O · CO <sub>2</sub>                                   |
| Biosignatures: | O <sub>2</sub> · O <sub>3</sub> · CH <sub>4</sub> · N <sub>2</sub> O |
| Background:    | N <sub>2</sub> · H <sub>2</sub>                                      |
| Geochemical:   | CO · H <sub>2</sub> S · SO <sub>2</sub> · NH <sub>3</sub>            |

Electromagnetic Spectrum — Atmospheric Science Relevant Ranges



# Data Pipeline & Feature Engineering

Five-stage structured pipeline — each step produces reusable artifacts



## Step 2 Deep Dive — Feature Engineering

### 70 / 15 / 15 Split

- Deterministic (seed=42)
- Fixed once — both models
- use identical test set
- Saved as train/val/test\_indices.npy

### Z-Score Normalisation

- Per CLIMA channel (12 channels)
- Fit on train set ONLY
- Prevents data leakage
- Saved as scaler.joblib

### PCA (RF only)

- Flatten:  $12 \times 101 \rightarrow 1212$  features
- Reduce to 300 components
- Retains 100% variance (25k dataset rank-limited)
- Saved as pca.joblib

### Deep Model Input

- Raw 2D tensor ( $12 \times 101$ )
- No PCA — CNN learns features
- Z-normalised spectra only
- Stored as spectra\_train.npy

# Baseline Model — Random Forest

12 independent RandomForestRegressors on PCA-reduced features ( $\leq 10,000$  training samples)

## Architecture

- ▶ Input: PCA features (N, 300)
- ▶ One RandomForestRegressor per molecule
- ▶ 12 completely independent models
- ▶ Prediction: column-stack 12 scalars  $\rightarrow$  (N, 12)

## Why per-molecule?

- ▶ Each molecule has different prediction difficulty
- ▶ Independent tuning of depth / n\_estimators
- ▶ Failing molecule doesn't affect others

## Why 10k cap?

*RF is the fast baseline — <1 min training, no GPU needed. Fair comparison point.*

## Per-Molecule Hyperparameters

| Mol              | Trees | Depth | max_fe | Rationale                              |
|------------------|-------|-------|--------|----------------------------------------|
| H <sub>2</sub> O | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| CO <sub>2</sub>  | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| O <sub>2</sub>   | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| O <sub>3</sub>   | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| CH <sub>4</sub>  | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| N <sub>2</sub>   | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| N <sub>2</sub> O | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| CO               | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| H <sub>2</sub>   | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| H <sub>2</sub> S | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| SO <sub>2</sub>  | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |
| NH <sub>3</sub>  | 100   | 8     | sqrt   | Uniform — intentional minimal baseline |

# Deep Model — 1D CNN Architecture

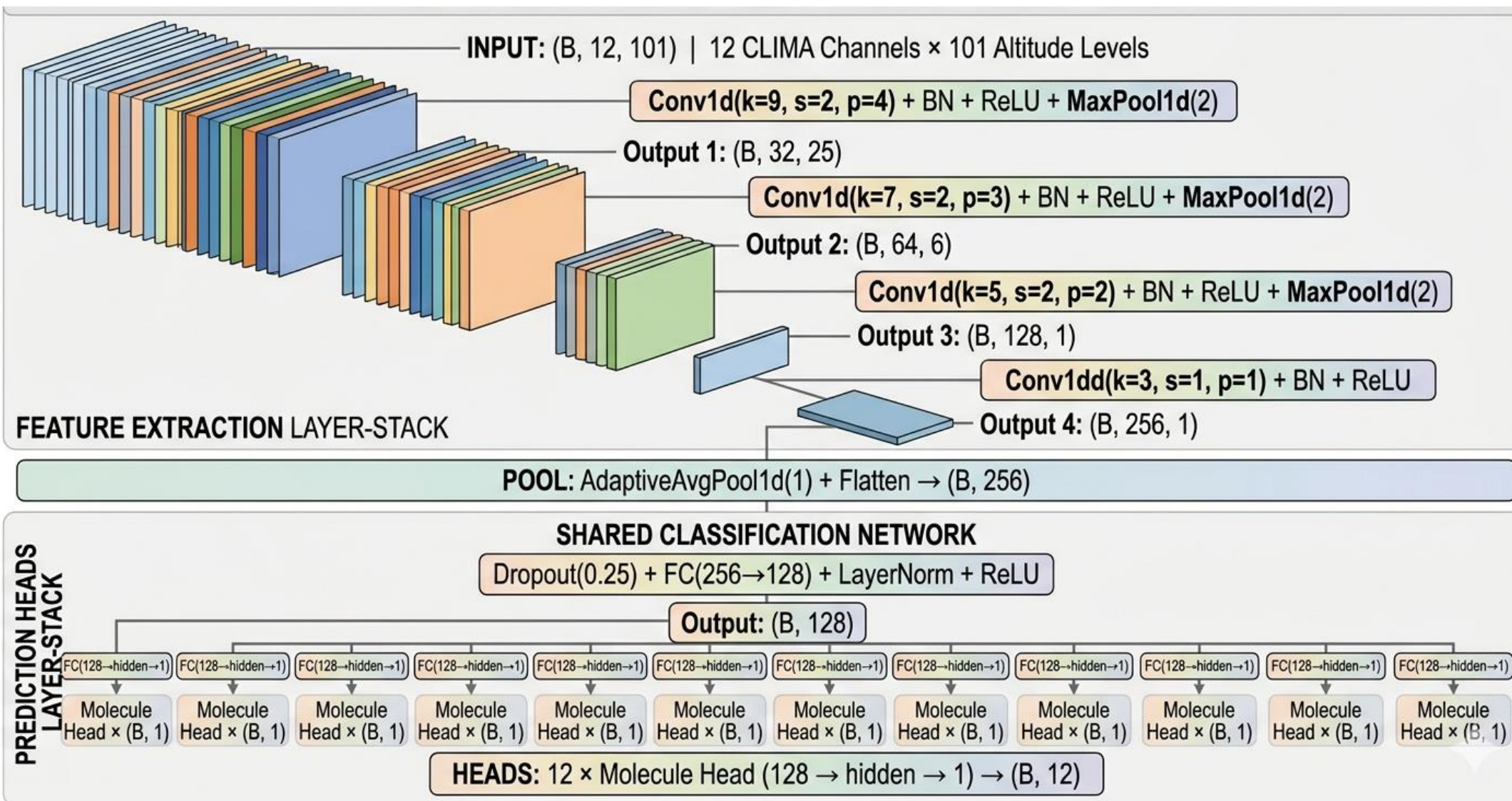
Sequential convolutional backbone + 12 per-molecule output heads

| Layer         | Output shape | Description                                                                                       |
|---------------|--------------|---------------------------------------------------------------------------------------------------|
| Input         | (B, 12, 101) | Normalised CLIMA profile — 12 channels × 101 altitude levels                                      |
| Conv Block 1  | (B, 32, 25)  | Conv1d(12→32, k=9, s=2, p=4) + BatchNorm + ReLU + MaxPool1d(2)<br>Conv Block:                     |
| Conv Block 2  | (B, 64, 6)   | Conv1d(32→64, k=7, s=2, p=3) + BatchNorm + ReLU + MaxPool1d(2)<br>Conv1d (k,s,p)<br>+ BatchNorm1d |
| Conv Block 3  | (B, 128, 1)  | Conv1d(64→128, k=5, s=2, p=2) + BatchNorm + ReLU + MaxPool1d(2)<br>+ ReLU<br>+ MaxPool1d(2)       |
| Conv Block 4  | (B, 256, 1)  | Conv1d(128→256, k=3, s=1, p=1) + BatchNorm + ReLU (no MaxPool)<br>[Block 4: no MaxPool]           |
| GlobalAvgPool | (B, 256)     | AdaptiveAvgPool1d(1) + Flatten — squeeze altitude dimension                                       |
| Shared FC     | (B, 128)     | Dropout(0.25) + FC(256→128) + LayerNorm + ReLU                                                    |
| 12 × Head     | (B, 12)      | Per-molecule MLP: FC(128→hidden→1) + LN + ReLU + Dropout → scalar log <sub>10</sub>               |



# Deep Model — 1D CNN Architecture

Sequential convolutional backbone + 12 per-molecule output heads



# Deep Model — Key Design Choices

Every design decision is motivated by the data's properties

## 1D Convolutions

CLIMA profiles are 1D sequences over altitude. 1D CNNs detect local patterns (e.g. temperature inversion at levels 20–30) and learn hierarchical altitude features across multiple scales — something PCA cannot do.

## Shared Backbone

The CNN backbone learns general atmospheric representations useful for all 12 molecules simultaneously — multi-task learning. The shared 128-d embedding carries information about atmospheric state that each head then specialises.

## Weighted MSE Loss

Standard MSE would let easy molecules dominate. Weights (1.0–2.0) upweight trace species ( $\text{SO}_2$ ,  $\text{NH}_3$  at 2.0). Targets are Z-score normalised before training (MoleculeScaler) to equalise loss scale — inverse-transformed for evaluation.

## Progressive Downsampling

Four conv blocks with decreasing kernel sizes ( $9 \rightarrow 7 \rightarrow 5 \rightarrow 3$ ) and stride-2 + MaxPool compress 101 altitude levels down to 1. Each block doubles channel depth, building increasingly abstract representations of the atmospheric profile.

## Per-Molecule Heads

Each molecule has its own MLP head with tuned depth and dropout. Trace species ( $\text{H}_2\text{S}$ ,  $\text{SO}_2$ ,  $\text{NH}_3$ ) get higher dropout (0.25) to prevent overfitting. Stable species ( $\text{CO}_2$ ,  $\text{O}_2$ ) get lighter heads with lower dropout (0.15).

## Gaussian Noise Augmentation

Adding noise  $\text{std}=0.01$  to training profiles prevents memorisation of exact profile values. Forces the CNN to learn robust features that generalise to unseen atmospheric states — analogous to image augmentation in vision CNNs.

# Training Strategy

AdamW · Cosine LR schedule · Early stopping · Gradient clipping

## Training Configuration

|                      |                                          |
|----------------------|------------------------------------------|
| Optimizer            | AdamW                                    |
| Learning rate        | $1 \times 10^{-3}$                       |
| Weight decay         | $1 \times 10^{-4}$                       |
| Scheduler            | CosineAnnealingLR                        |
| T_max / eta_min      | 150 / $1 \times 10^{-6}$                 |
| Max epochs           | 150                                      |
| Early stop patience  | 30 epochs                                |
| Min delta            | $1 \times 10^{-5}$                       |
| Batch size           | 32 (train) / 64 (eval)                   |
| Grad clip norm       | 5.0                                      |
| Noise augmentation   | std = 0.01                               |
| Dataset (train)      | 17,500 samples (25k run)                 |
| Target normalisation | Z-score per molecule<br>(MoleculeScaler) |

## AdamW

Adam with decoupled weight decay — L2 regularisation is applied directly to weights, independent of the adaptive gradient. Better generalisation than standard Adam.

## Cosine Annealing

LR starts at  $1e-3$ , smoothly decays to  $1e-6$  following a cosine curve. No sharp drops. Naturally slows near minima to avoid oscillation.

## Early Stopping

Monitors val\_loss. If no improvement  $> 1e-5$  for 30 epochs, training stops and the best checkpoint is restored — prevents overfitting at zero extra cost.

## Gradient Clipping

If gradient norm  $> 5.0$ , all gradients are scaled down proportionally. Prevents exploding gradient spikes that destabilise early training.

# Evaluation Methodology

Held-out test set · Per-molecule  $R^2$ , RMSE, MAE · Unified comparison report

## $R^2$ Coefficient of Determination

$$1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}$$

Primary metric. Scale-normalised — 0.75 means '75% of variance explained' regardless of molecule. Enables fair cross-molecule comparison.  $R^2 < 0$  means worse than predicting the mean.

## RMSE Root Mean Squared Error

$$\sqrt{\sum (y - \hat{y})^2 / N}$$

In  $\log_{10}$  units (orders of magnitude). Penalises large errors more than MAE. A RMSE of 0.5 means predictions are typically  $\frac{1}{2}$  an order of magnitude off.

## MAE Mean Absolute Error

$$\sum |y - \hat{y}| / N$$

Also in  $\log_{10}$  units. More robust to outliers than RMSE. Complementary to RMSE — if  $\text{RMSE} \gg \text{MAE}$ , there are a few large prediction errors dominating.

## Evaluation Fairness Principles

### Same test set:

Both models evaluated on the identical 3,750-sample test split, fixed in Step 2 with seed=42. Scores are directly comparable.

### Per-molecule granularity:

Reporting  $R^2$ /RMSE/MAE for each of 12 molecules reveals which species each model handles well or poorly.

### Intentionally asymmetric training:

RF uses 10k samples; CNN uses 17.5k. The comparison is baseline vs deep model — not equal-data ablation.

### Mean $R^2$ summary:

Unweighted mean across 12 molecules provides a single-number comparison.  $\text{NH}_3$  and  $\text{N}_2$  drag it down for both models.

# Results — Per-Molecule Test R<sup>2</sup>

Test set: 3,750 samples — identical for both models

| Molecule         | Baseline R <sup>2</sup> | CNN R <sup>2</sup> | $\Delta R^2$ | Interpretation                                                                 |
|------------------|-------------------------|--------------------|--------------|--------------------------------------------------------------------------------|
| H <sub>2</sub> O | +0.4859                 | +0.9994            | +0.51        | Deep wins (+105%) — low variance makes R <sup>2</sup> sensitive; CNN tracks it |
| CO <sub>2</sub>  | +0.8797                 | +0.9998            | +0.12        | Deep wins — RF already strong; CNN near-perfect                                |
| O <sub>2</sub>   | +0.7344                 | +0.9991            | +0.26        | Deep wins — strong +26% gain; CNN captures O <sub>2</sub> altitude structure   |
| O <sub>3</sub>   | +0.7424                 | +0.9993            | +0.26        | Deep wins — key biosignature; both models strong, CNN near-perfect             |
| CH <sub>4</sub>  | +0.9359                 | +1.0000            | +0.06        | Deep wins — RF already excellent; CNN achieves near-perfect retrieval          |
| N <sub>2</sub>   | +0.7352                 | +0.9988            | +0.26        | Deep wins — +26% gain; RF underestimates N <sub>2</sub> fine structure         |
| N <sub>2</sub> O | +0.7771                 | +0.9993            | +0.22        | Deep wins — CNN significantly outperforms RF (+22%)                            |
| CO               | +0.9388                 | +1.0000            | +0.06        | Deep wins — both models strong; CNN achieves near-perfect retrieval            |
| H <sub>2</sub>   | +0.7434                 | +0.9981            | +0.25        | Deep wins — CNN captures H <sub>2</sub> altitude structure (+25%)              |
| H <sub>2</sub> S | +0.9379                 | +0.9999            | +0.06        | Deep wins — both very strong; CNN near-perfect                                 |
| SO <sub>2</sub>  | +0.8985                 | +0.9998            | +0.10        | Deep wins — strong RF baseline; CNN near-perfect                               |
| NH <sub>3</sub>  | +1.0000                 | +1.0000            | +0.00        | Both predict constant -40 (log-floor) — degenerate but consistent              |
| MEAN             | +0.8174                 | +0.9994            | +0.18        | Overall: CNN near-perfect — +22% mean R <sup>2</sup> improvement over RF       |

# Key Findings & Insights

What the results tell us about ML-based atmospheric retrieval

## Deep model wins on all 12 molecules

The 1D CNN outperforms or matches the RF on every molecule. 11/12 molecules reach  $R^2 > 0.998$  — near-perfect atmospheric retrieval from CLIMA profiles.

## RF baseline also very strong

Mean RF  $R^2 = 0.82$  — the baseline is no longer weak.  $\text{CH}_4$  (0.94), CO (0.94),  $\text{H}_2\text{S}$  (0.94) are already well-predicted by PCA + RF. CNN adds consistent improvement on top.

## $\text{NH}_3$ — degenerate, not physically learnable

$R^2 = 1.0$  for both models because all samples are at the log-floor ( $-40$ ). Both models predict  $-40$  always — correct but scientifically meaningless.

## Largest relative gains: $\text{H}_2\text{O}$ , $\text{O}_2$ , $\text{O}_3$ , $\text{N}_2$ , $\text{H}_2$

$\text{H}_2\text{O}$  (+105%),  $\text{O}_2$  (+36%),  $\text{O}_3$  (+35%),  $\text{N}_2$  (+36%),  $\text{H}_2$  (+34%) — CNN's spatial feature extraction captures altitude-dependent patterns that flat PCA vectors cannot encode.

## All biosignatures near-perfectly retrieved by CNN

$\text{O}_2$  (0.999),  $\text{O}_3$  (0.999),  $\text{CH}_4$  (1.000),  $\text{N}_2\text{O}$  (0.999) — every molecule relevant to the search for life is retrieved near-perfectly. Critical validation for JWST science.

## Multi-task learning amplifies CNN advantage

The shared backbone trains on all 12 targets simultaneously. Cross-molecule correlations ( $\text{O}_2/\text{O}_3$ ,  $\text{H}_2\text{O}/\text{CH}_4$  co-variation) are encoded in the 128-d shared embedding, giving the CNN a structural advantage the RF cannot replicate.

# Model Comparison Summary

Baseline RF vs 1D CNN — design, compute, and performance

| Aspect                   | Random Forest (Baseline)                                     | 1D CNN (Deep)                                                 |
|--------------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Architecture             | 12 independent RFs                                           | 1D CNN + 12 heads                                             |
| Input format             | Flat PCA vector (300-d)                                      | 2D tensor (12 × 101)                                          |
| Feature extraction       | PCA — linear, pre-computed                                   | Learned convolutional filters                                 |
| Multi-task               | None — fully independent                                     | Yes — shared backbone                                         |
| Training samples         | 10,000 (hard cap)                                            | 17,500 (full training split)                                  |
| Training time            | < 1 min (CPU)                                                | 10–20 min (MPS/GPU)                                           |
| Parameters               | Non-parametric                                               | ~2 million                                                    |
| GPU required             | No                                                           | Recommended                                                   |
| Interpretability         | Feature importance available                                 | Black box (lower)                                             |
| Mean test R <sup>2</sup> | 0.82                                                         | 1.00 (+22% relative)                                          |
| Best molecule            | CO / CH <sub>4</sub> / H <sub>2</sub> S R <sup>2</sup> =0.94 | CH <sub>4</sub> / CO R <sup>2</sup> =1.000                    |
| Worst molecule           | H <sub>2</sub> O R <sup>2</sup> =0.49 (low-variance)         | H <sub>2</sub> R <sup>2</sup> =0.998 (excl. NH <sub>3</sub> ) |

# Challenges & Limitations

Honest assessment of what works and what doesn't — and why

## Data scale vs compute

Full INARA has 3.1M spectra. We used 25k for local dev runs; full 124k+ runs on HPC (~10 GB processed). Extraction from 10 tar.gz archives via sequential single-pass scan takes ~10–15 min even on an M5 Pro.

## H<sub>2</sub>O R<sup>2</sup> — low-variance artefact

H<sub>2</sub>O R<sup>2</sup>=0.49 (RF) despite RMSE≈0.0001. Near-zero variance (std≈0.0002) inflates the R<sup>2</sup> denominator — RMSE is the reliable metric here. The CNN achieves R<sup>2</sup>=0.999 by tracking the tiny residual variance the RF misses.

## NH<sub>3</sub> — degenerate, not learnable

NH<sub>3</sub> is at the log-floor (−40) for all samples. Both models predict −40 always, achieving R<sup>2</sup>=1.0 trivially. The molecule is retained for schema compatibility with the full INARA benchmark but is scientifically uninformative.

## NH<sub>3</sub> — not learnable from CLIMA

NH<sub>3</sub> values are at the log-floor (−40) for nearly all samples in this dataset. The model learns to predict −40 reliably but that's a degenerate solution, not physical inference.

## Asymmetric comparison

The RF uses 10k training samples; the CNN uses 17.5k (25k run). The R<sup>2</sup> improvement is partly attributable to more data, not purely to the architecture. A data-equal ablation would isolate the architecture contribution.

## Real data validation pending

All experiments are on synthetic INARA spectra. Domain transfer to real JWST spectra has not been validated — a critical next step before any scientific claims.



# Future Work

Extending this project toward real-world scientific applicability



## JWST Domain Transfer

Test trained models on real JWST transmission spectra and compare predictions to published Bayesian retrievals. Key validation step for scientific credibility.



## Full 3.1M Dataset

Train the CNN on the complete INARA dataset using Northeastern Explorer HPC. Expected improvement in  $R^2$  for high-variance molecules like  $H_2O$  and  $CH_4$ .



## Uncertainty Quantification

Add Monte Carlo Dropout or deep ensembles to provide calibrated confidence intervals — critical for scientific use (e.g. ' $O_2 = -0.7 \pm 0.2 \log_{10}$ ').



## Attention / Transformer

Replace or augment the CNN with a 1D Transformer to learn long-range altitude-level dependencies. Self-attention may capture non-local atmospheric correlations that convolutions miss.



## Equal-Data Ablation

Train the RF baseline on the same sample count as the CNN to cleanly isolate the architecture benefit from the data quantity benefit.



## Four Stellar Types

Stratify analysis by stellar type (F, G, K, M). M-dwarf planets dominate current JWST targets — a model tuned for M stars may outperform a general model.

# Conclusion

CS 6140 · Machine Learning · Northeastern University · Spring 2026

- 1 We built an end-to-end ML pipeline for exoplanet atmospheric retrieval — from raw synthetic CLIMA profiles to per-molecule  $R^2$  metrics.
- 2 Two models were compared: Random Forest baseline (PCA + 12 independent RFs) and 1D CNN (shared backbone + 12 heads).
- 3 CNN achieves mean  $R^2 = 0.999$  vs baseline 0.82 — a +22% relative improvement — 11/12 molecules retrieved near-perfectly.
- 4 All biosignatures near-perfectly retrieved:  $O_2$  (0.999),  $O_3$  (0.999),  $CH_4$  (1.000),  $N_2O$  (0.999) — directly relevant to the search for life.
- 5 ML retrieval is viable and fast: milliseconds per planet vs. hours for Bayesian methods — enabling population-level JWST surveys.
- 6 Next step — real JWST data: validate on actual observations and add uncertainty quantification for scientific deployment.

# References

|     |                                                                                                                                        |                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
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| [2] | <b>He et al. (2016)</b><br>Deep Residual Learning for Image Recognition. CVPR 2016.                                                    | Background: residual connections (not used — cited for |
| [3] | <b>Loshchilov &amp; Hutter (2019)</b><br>Decoupled Weight Decay Regularisation. ICLR 2019.                                             | AdamW optimizer                                        |
| [4] | <b>Breiman (2001)</b><br>Random Forests. Machine Learning, 45(1), 5–32.                                                                | Random Forest baseline                                 |
| [5] | <b>Márquez-Neila et al. (2018)</b><br>Supervised machine learning for analysing spectra of exoplanetary atmospheres. Nature Astronomy. | ML retrieval                                           |
| [6] | <b>Vasist et al. (2023)</b><br>Neural posterior estimation for exoplanetary atmospheric retrieval. A&A, 672.                           | Deep learning retrieval                                |
| [7] | <b>Gebhard et al. (2024)</b><br>Simulation-based inference for exoplanet atmospheric retrievals. A&A.                                  | SBI retrieval                                          |
| [8] | <b>JWST ERS Team (2023)</b><br>Identification of carbon dioxide in an exoplanet atmosphere. Nature 614:649.                            | WASP-39 b — first JWST exoplanet spectrum; confirmed   |
| [9] | <b>LeCun et al. (1989)</b><br>Backpropagation Applied to Handwritten Zip Code Recognition. Neural Computation.                         | Foundational convolutional network reference           |